Learning When Black Hole Hair Is Observable: Physical Identifiability, Generalization, and Observable Complementarity

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Abstract

Black hole spectroscopy seeks to infer spacetime parameters from observable signals, but accurate recovery on simulations does not guarantee that those parameters are physically distinguishable. We study this problem in a two-parameter Kiselev spacetime, whose Schwarzschild limit at $k = 0$ provides an exact non-identifiable control. Across an 11×11 physical grid, we combine leading-eikonal ringdown quantities with redshift, timing, and photon-geometry features from timelike trajectories and three-dimensional null-geodesic shooting. Adding photon geometry increases the median minimum Jacobian singular value by $4.54\times$ under the declared normalization; redshift also supplies complementary parameter information. In grouped tests, the normalized mean absolute error for identifiable w_q decreases from 0.309 for ringdown alone to 0.087 with photon geometry. A non-learned inverse baseline supports the qualitative improvement. Importantly, the complementarity persists when the analysis is restricted to observation-facing quantities, using only leading-eikonal Ω and λ, redshift, and signed sky coordinates, without latent photon-orbit descriptors as inputs. The forward calculation passes targeted convergence tests, but tree estimators remain sensitive in their prediction tails and the MLP is unstable under directional extrapolation. These findings establish observable complementarity in a controlled theoretical benchmark, not detector-level parameter constraints.

1 Introduction

Black hole spectroscopy aims to infer properties of a compact object's spacetime from observable signals, such as quasinormal-mode frequencies and damping rates [8, 2, 3]. In practice, however, accurate recovery of model parameters from simulated data does not necessarily mean that those parameters are physically distinguishable. Different parameter combinations can produce nearly identical observables, and random data partitions can further obscure this problem by mixing closely related physical systems between training and test sets.

We study this issue in a two-parameter Kiselev spacetime [12]. In the metric considered here, k sets the overall amplitude of the deviation from Schwarzschild, while w_q controls how that deformation scales with radius. When $k = 0$, the deformation vanishes entirely and the spacetime reduces exactly to Schwarzschild. In that limit, all dependence on w_q disappears, so no physical observable can contain information about w_q. This provides an analytically known case of exact non-identifiability against which the inference procedure can be tested.

Away from the Schwarzschild limit, the central question is whether the available observables respond to k and w_q in sufficiently different ways to separate them. We therefore combine leading-eikonal ringdown quantities with observables obtained from timelike emitter motion and three-dimensional null-geodesic shooting. These include photon redshift, timing information, impact parameters, and signed observer-sky coordinates. The aim is not simply to ask whether these quantities change when the spacetime changes, but whether they provide independent parameter information.

We quantify this using the Jacobian of the forward observable map. Its singular values measure how strongly different parameter directions are resolved locally, while its condition number diagnoses how easily finite perturbations can be amplified in the inverse problem. We complement this local analysis with machine-learning and non-learned inverse models evaluated under random interpolation, grouped physical holdouts, and directional extrapolation. Each physical spacetime contributes one inference sample, so repeated phase samples from the same system never cross training and test partitions.

Our main result is that leading-eikonal ringdown alone leaves a weakly constrained parameter direction, whereas photon geometry and redshift provide complementary information. Photon geometry gives the larger improvement in the weakest local Jacobian direction, and both channels substantially improve recovery of identifiable w_q values under grouped physical holdouts. Importantly, this improvement persists when latent geometric descriptors are removed: a restricted analysis using only Ω, λ, redshift, and signed sky coordinates still yields substantially better identifiability and inverse recovery. These gains are also reproduced by a non-learned nearest-model inverse, showing that they are not specific to a particular machine-learning architecture.

This study is therefore a controlled identifiability benchmark rather than a detector-level inference analysis. The goal is to determine what information the forward observables contain before optimizing an inverse estimator. We distinguish structural non-identifiability, where different parameter values are physically indistinguishable, from practical identifiability, where the parameters are formally distinct but difficult to recover at finite precision. This distinction also connects to identifiability analysis in dynamical inverse problems [20, 22].

The study complements neural posterior-estimation approaches [7, 11, 10, 6, 17, 16] by asking what information the selected forward observables contain before an inverse estimator is chosen. The ringdown framework builds on Ref. [25]; the new elements here are the geodesic-shooting observables, local identifiability diagnostics, leakage-aware learning protocols, non-learned baselines, observation-facing sensitivity tests, and numerical robustness audits.

2 Physical Model and Identifiability Framework

The static spherical line element is

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \tag{1}$$

$$f(r) = 1 - \frac{2M}{r} - \frac{k}{r^{1+3w_q}}. \tag{2}$$

Here k sets the overall amplitude of the deviation from Schwarzschild, while w_q controls how that deviation scales with radius; we use geometrized units with $M = 1$. Leading-eikonal ringdown features use the unstable circular null orbit, with the limitations of the geodesic–QNM correspondence kept explicit [5, 14, 25].

We evaluate an 11×11 analysis grid: 121 systems with $k \in [0, 0.0025]$ and $w_q \in [-0.7125, -0.45]$. The $k = 0$ systems are retained deliberately because their Schwarzschild metric is independent of

w_q and therefore supplies the exact structural non-identifiability boundary. Each grid point denotes one physical system; its full phase-resolved simulation is later compressed into one feature vector.

For standardized observables $\tilde{\mathbf{O}}$ and parameters $\boldsymbol{\theta} = (k, w_q)$, the local Jacobian is

$$J_{ij} = \frac{\partial \tilde{O}_i}{\partial \theta_j}. \tag{3}$$

The fixed dimensionless row is $J_i = (\Delta k\, \partial_k O_i/s_i, \Delta w_q\, \partial_{w_q} O_i/s_i)$, where s_i is the global q95–q05 range over the accepted 121-point nominal grid and $\Delta k = 0.0025$ and $\Delta w_q = 0.2625$ are the full science-domain spans. Features with non-finite values or $s_i \leq 10^{-12}$ are excluded rather than divided by the floor. These choices fix the units of the SVD and therefore condition the numerical values of $\sigma_{\min}$ and $\kappa(J)$. Higher $\sigma_{\min}(J)$ and lower $\kappa(J) = \sigma_{\max}/\sigma_{\min}$ are favorable. At $k = 0$, the deformation term vanishes, so $f(r)$ is Schwarzschild and independent of w_q. Every physical observable is therefore independent of w_q, the w_q column of J vanishes, and the two-parameter Jacobian has rank one with $\sigma_{\min} = 0$ in the analytic limit. We treat this chain as a ground-truth null control, not merely an edge case: any inferred physical w_q information there would be spurious. Nonzero rank away from that boundary does not by itself imply practical recoverability at finite precision. Exact $k = 0$ rank loss is structural, while small nonzero k can be full-rank yet practically ill-conditioned. Historical synthetic and proxy results are kept separate in Appendix A. Having defined the inverse problem and its exact rank-deficient boundary, we next construct the physical observables used to evaluate it.

3 Physical Geodesic Shooting and Numerical Validation

For each analysis-grid pair (k, w_q), we construct the Kiselev spacetime and integrate one timelike emitter orbit with $M = 1$, $r_p = 8M$, $r_a = 12M$, and physical coordinate azimuth $\phi \in [\pi, 3\pi]$. The emitter starts at apocentre at $\phi = \pi$. We then select a finite set of emission phases. At each phase we read the emitter state, solve the two launch angles of the direct photon branch, integrate the null ray to the observer at $(0, 0, -80M)$, and project its momentum into the static observer tetrad. The nominal feature archive uses 81 phase samples per system; the uniform 161-phase and targeted 321-phase calculations are numerical audits discussed later, not replacements for the nominal archive.

The emitter follows the equatorial timelike Hamiltonian

$$H = \tfrac{1}{2}[-E^2/f + f p_r^2 + L^2/r^2] = -\tfrac{1}{2}. \tag{4}$$

Its physical coordinate azimuth begins at apocentre: $\phi = \pi$, $r = r_a$, $p_r = t = \tau = 0$. Coordinate ϕ is not treated as a radial anomaly; turning points are detected from $p_r = 0$ and the radial azimuthal period includes apsidal advance.

Photons are propagated in Cartesian spatial coordinates with covariant momentum $\mathbf{p}$ and $p_t = -1$,

$$H_\gamma = \tfrac{1}{2}[-1/f + \mathbf{p}^2 + (f-1)(\mathbf{n}\cdot\mathbf{p})^2] = 0. \tag{5}$$

Two launch angles are solved so the direct branch reaches the observer plane. The implementation is designed to report any failed phase rather than interpolate or replace it with a synthetic geodesic proxy. In the accepted 121-system science grid, the required shooting-success fraction is unity. Photon momenta are projected onto the static observer tetrad before sky angles are formed. At every successful phase we record redshift, impact parameter, signed observer-tetrad sky coordinates, excess delay, and relative arrival time. Redshift uses $1 + z = \omega_{\rm emit}/\omega_{\rm obs}$, and direct arrival time is

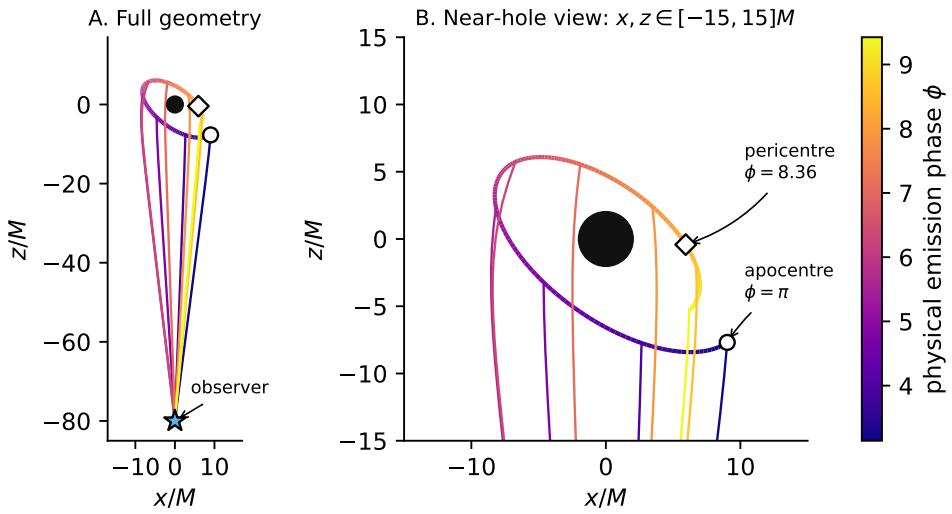


Figure 1: Representative direct-branch null geodesics for $M = 1$, $r_p = 8M$, $r_a = 12M$, $k = 10^{-3}$, $w_q = -0.5$, and observer $(0, 0, -80M)$. Physical phase runs from π at apocentre to 3π; the diamond marks the numerically detected pericentre and the dark disk is a horizon-scale marker, not a shadow. At each phase, two launch angles are solved until the ray reaches the observer. The right panel shows the stated near-hole coordinate bounds on their actual scale; no trajectory is geometrically magnified. This is a controlled dimensionless benchmark.

compared with $\int(1+z)d\tau_{\rm emit}$ after removing one common additive constant. For clarity, the reported propagation time is $t_{\rm prop} = t_{\rm hit} - t_{\rm emit}$, and the excess delay removes the corresponding Euclidean source–observer distance, $\Delta t_{\rm excess} = t_{\rm prop} - \|\mathbf{x}_{\rm obs} - \mathbf{x}_{\rm emit}\|$. Relative arrival time is expressed on the static observer's clock as $T_{\rm rel} = \sqrt{f_{\rm obs}}[(t_{\rm emit} + t_{\rm prop}) - (t_{\rm emit} + t_{\rm prop})_{\rm ref}]$. The separately reported $T_z = \int_{\tau_{\rm ref}}^{\tau}(1+z)\,d\tau_{\rm emit}$ (`toa_from_redshift`) is an independent validation quantity obtained by cumulative trapezoidal integration, not a replacement for $T_{\rm rel}$.

An independent implementation cross-checked the emitter and photon dynamics, observer-frame observables, extracted features, and selected Jacobians without importing production `bhhairml` physics code. The maximum trajectory/observable, full feature-vector, and selected-Jacobian absolute differences were 7.63×10^{-9}, 7.77×10^{-11}, and 2.62×10^{-13}, respectively.

In the Schwarzschild benchmark, all 161 phases succeed for both tested nominal w_q values, which agree because $k = 0$. The independent radial benchmark validates the turning-point period and apsidal advance. The physical 121-point grid also preserves the Hamiltonian constraints and exact rank-loss boundary. The overlapping arrival-time phase-resolved series and their numerical residuals are retained as supporting evidence in Appendix H and Fig. 15.

Figure 1 makes the direct-branch geometry explicit using actual validated emitter states, launch angles, photon integrations, and observer geometry. It is a controlled benchmark visualization, not a radiative-transfer or detector simulation.

Figure 2 shows examples of the raw observables recorded at the selected phases for three validated spacetimes.

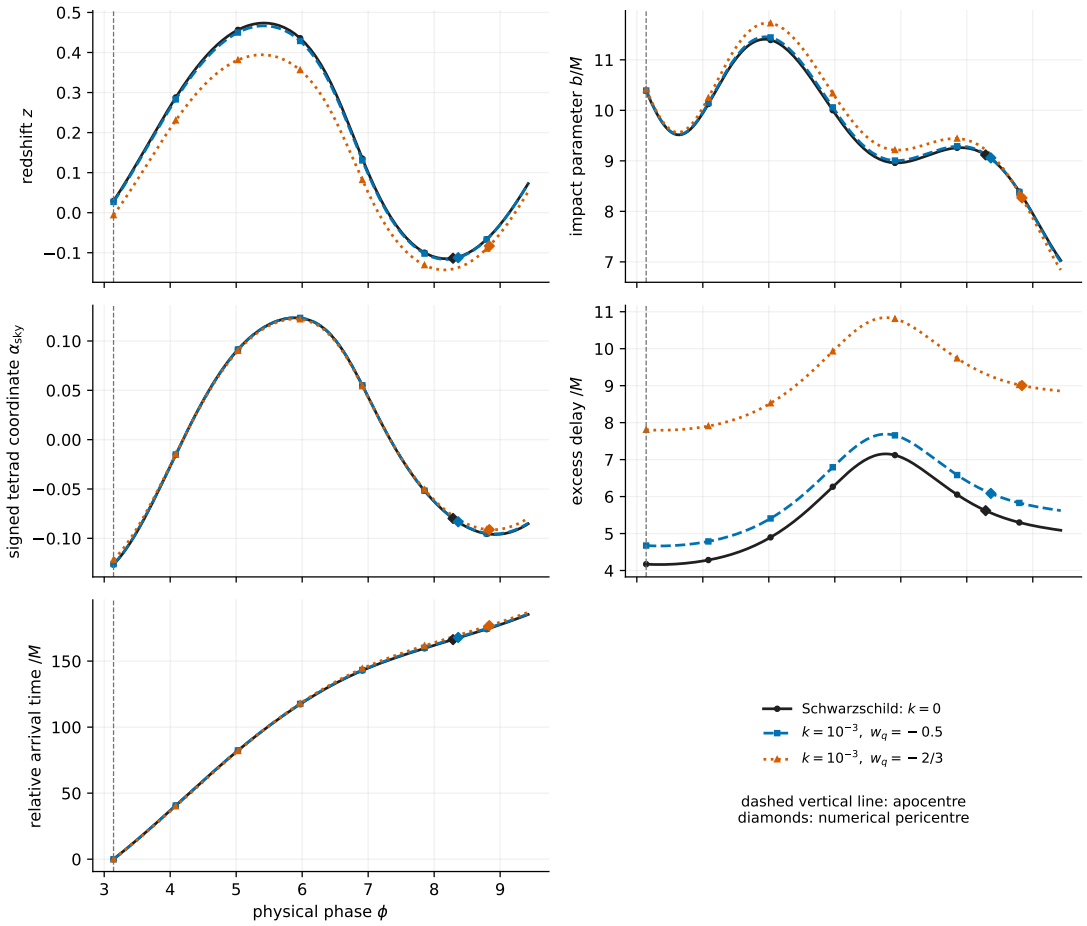


Figure 2: Physical direct-branch observables respond differently to the same spacetime parameters. They are shown over $\phi \in [\pi, 3\pi]$ for Schwarzschild and $k = 10^{-3}$ with $w_q = -0.5, -2/3$, using $M = 1$, $r_p = 8M$, $r_a = 12M$, and observer $(0, 0, -80M)$. The dashed line marks the initial apocentre and diamonds mark numerical pericentres. All phase-resolved series use the same successful matched phases without interpolation. Observable response amplitude and independent parameter information are not equivalent. Markers highlight directly simulated phase samples; connecting lines are visual guides only and do not represent interpolated physical values.

These discrete phase-resolved samples are the quantities subsequently compressed into fixed-dimensional feature vectors.

3.1 From phase-resolved observables to feature vectors

For one physical system and one raw observable, the solver returns the finite sequence $O(\phi_1), O(\phi_2), \ldots, O(\phi_N)$ at the selected emission phases. These are directly simulated samples, not an interpolated continuous curve. We summarize each sampled phase series by its mean, amplitude, reference value where defined, and sine/cosine coefficients through the third harmonic. The Fourier coefficients compress the successful computed samples; they are never used to fill a failed phase.

The feature families are defined before any comparison: *ringdown* contains four leading-eikonal scalar features; *orbital* contains the radial azimuthal period, apsidal advance, and summaries of r and p_r (20 features); *photon geometry* summarizes impact parameter and the two signed observer-tetrad sky coordinates (27); *redshift* contains 9 summaries; and *timing* summarizes excess delay and relative arrival time (17). *All shooting* is orbital plus photon geometry, redshift, and timing (73). Ringdown plus photon geometry has 31 features, and ringdown plus all shooting has 77. Appendix C gives the exact predefined table.

4 Physical Observable Complementarity

A large response can reveal that the spacetime changed without revealing which parameter caused the change. Timing can remain nearly collinear in parameter space, whereas both redshift and photon geometry add complementary information to ringdown. Photon geometry gives the larger improvement in the weakest local direction, while redshift provides a substantial complement that is more directly connected to a measurable spectroscopic quantity.

Large response is insufficient when the two parameter-response directions are nearly collinear. In Fig. 3, $\sigma_{\max}$ measures the strongest response, $\sigma_{\min}$ the weakest identifiable direction, their angle measures directional separation, and $\kappa(J) = \sigma_{\max}/\sigma_{\min}$ measures conditioning. The finite-difference diagnostic is anchored at $(k, w_q) = (10^{-3}, -0.5)$: the k response uses the step from Schwarzschild to that anchor, while the w_q response uses $-0.5 \to -2/3$ at fixed $k = 10^{-3}$.

The full observer tracks nearly overlap for these small deformations, but their matched-phase differences are numerically resolved; the actual-scale tracks and residuals are provided in Appendix H.

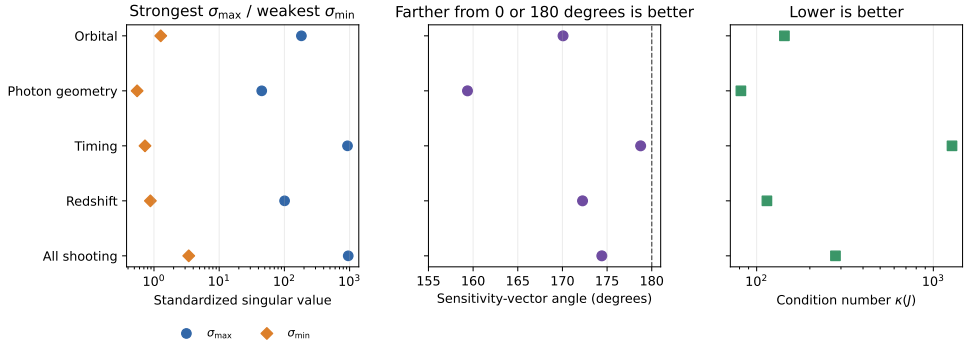


Figure 3: Local parameter-response diagnostics at $(k, w_q) = (10^{-3}, -0.5)$. Timing is strongly responsive but poorly conditioned ($\kappa(J) = 1275.5$); photon geometry improves conditioning ($\kappa(J) = 81.4$). The finite-difference stencil is defined in the text.

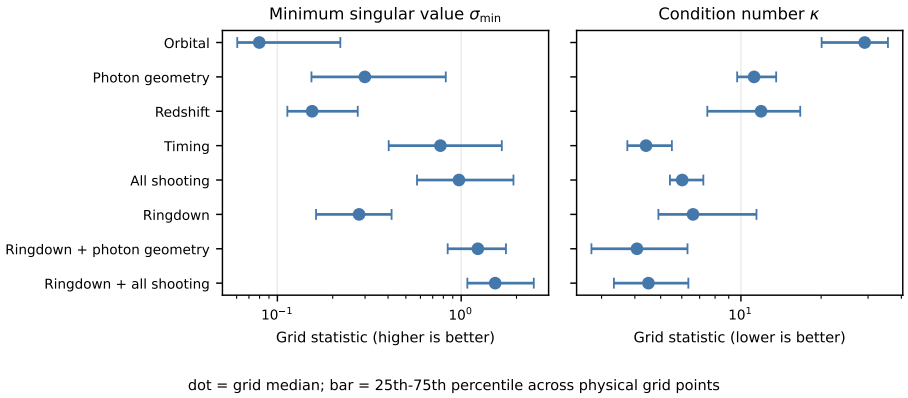


Figure 4: Nominal 81-phase grid-wide summaries of $\sigma_{\min}$ and condition number: each dot is the grid median and each bar spans the 25th–75th percentile across physical grid points, not measurement uncertainty. Higher $\sigma_{\min}$ and lower condition number are favorable. The separate 161-phase audit is reported in the text.

Table 1: Selected complementarity metrics comparing ringdown alone with two independently motivated additions: redshift and photon geometry. Jacobian medians use the nominal 81-phase archive; grouped and directional NMAE use the validated 161-phase comparison with predefined splits and score identifiable w_q only.

Feature set	med. $\sigma_{\min}$	med. $\kappa(J)$	grouped w_q NMAE	directional w_q NMAE
Ringdown	0.2782	6.6112	0.3093	0.3011
Rdown + redshift	0.9459	3.6107	0.0991	0.1922
Rdown + photon geom.	1.2326	4.0731	0.0865	0.2048

Across the 110 finite-k systems, adding photon geometry to ringdown raises the median pointwise minimum singular value by 4.54× and improves the median pointwise condition number by 2.28×. Redshift independently improves these quantities by 3.29× and 2.27×, respectively. Thus photon geometry produces the larger gain in the weakest identifiable direction, while redshift achieves slightly better median conditioning. The combined-grid medians are $(\sigma_{\min}, \kappa(J)) = (1.2326, 4.0731)$ for photon geometry and $(0.9459, 3.6107)$ for redshift. These gains are pointwise ratio medians, not ratios of grid medians.

The qualitative complementarity survives alternative observable scalings, although the numerical gain factors depend on normalization. The finite-grid bootstrap intervals, scaling comparisons, all-shooting results, and 161-phase cross-checks are documented in Appendix H.3 and Appendix H. Redshift is closer to a directly measurable spectroscopic channel than the idealized photon-geometry variables; this distinction matters when translating the benchmark into an observational model.

Does the representative complementarity seen at the reference point persist across the physical parameter grid? Figure 4 gives the primary grid-wide comparison. The pointwise maps show that complementarity varies across the finite-k domain; the full maps are provided in Appendix E.

4.1 Finite-domain ambiguity check

Because local Jacobian rank cannot rule out distant observable near-collisions, we also checked the sampled physical grid. No unresolved distant finite-k observable collision was found on the

121-point grid at the tested numerical resolution, and photon geometry improves the distant-pair separation diagnostics. This finite-grid check does not prove continuous global injectivity; the complete thresholds and neighbor diagnostics appear in Appendix I.

5 Leakage-Aware Inverse-Learning Protocols

Each physical system contributes one supervised inference sample: a single feature vector $\mathbf{x}$ constructed from its entire phase-resolved simulation, with target (k, w_q). Individual phase samples are not treated as independent training examples. Training, calibration, and test partitions are therefore formed at the physical-system level.

Five primary feature sets are ordered consistently: ringdown, photon geometry, all shooting, ringdown plus photon geometry, and ringdown plus all shooting. Here photon geometry and the other integrated quantities are physical shooting observables, distinct from the historical synthetic geodesic proxies. Models are histogram gradient boosting (HGB), random forest (RF), and a target-scaled multilayer perceptron (MLP) [4, 9, 18]. These heterogeneous standard estimators are retained intentionally rather than optimizing one high-capacity architecture: tree models expose piecewise split-crossing behavior, while the MLP supplies a smoother perturbation response and a distinct extrapolation failure mode. Their purpose is to reveal estimator-specific behavior, not to claim that standard models are universally preferable. The increasingly demanding protocols are random interpolation, two blocked 3×3 grouped physical layouts, and four directional extrapolation tests. All nominal $k = 0$ records are retained and flagged. They contribute to k scoring, but ordinary w_q error excludes them because w_q is exactly non-identifiable there. Although their nominal w_q labels differ, these 11 records form one physical Schwarzschild equivalence class. A collapsed-equivalence sensitivity analysis is reported below and in Appendix F.6.

Normalized mean absolute error divides MAE by the full target span. Split-conformal intervals target 90% coverage and are interpreted under their exchangeability assumptions [23, 21, 1]. Feature-noise scales and every preprocessing transform are learned on training folds. The 81-to-161 and targeted 161-to-321 comparisons apply archived frozen estimators; they do not retrain on higher-resolution test features.

6 Inverse-Learning Results

The secondary 161-phase NMAE values in Table 1 are used only for the matched redshift comparison and do not replace the primary protocol summary reported below. Random splitting is optimistic. For HGB, RF, and MLP considered together, random ringdown NMAE is 0.0361/0.1688 for k/identifiable w_q, while grouped ringdown is 0.0485/0.3093. Ringdown plus photon geometry gives grouped 0.0457/0.0875, with the w_q improvement reproduced independently in both tree families. Directional extrapolation remains harder: the combined set has aggregate NMAE 0.1305/0.2012. Table 1 isolates identifiable-w_q recovery for the 161-phase redshift comparison under the same predefined splits. Both added channels greatly improve grouped recovery relative to ringdown (0.3093): redshift reaches 0.0991, while photon geometry is slightly better at 0.0865. Under directional extrapolation, redshift (0.1922) and photon geometry (0.2048) both improve on ringdown (0.3011) and are comparable; neither channel is universally superior from these aggregate values. Figure 5 asks how recoverability changes as validation moves from random interpolation to grouped physical holdouts and directional extrapolation.

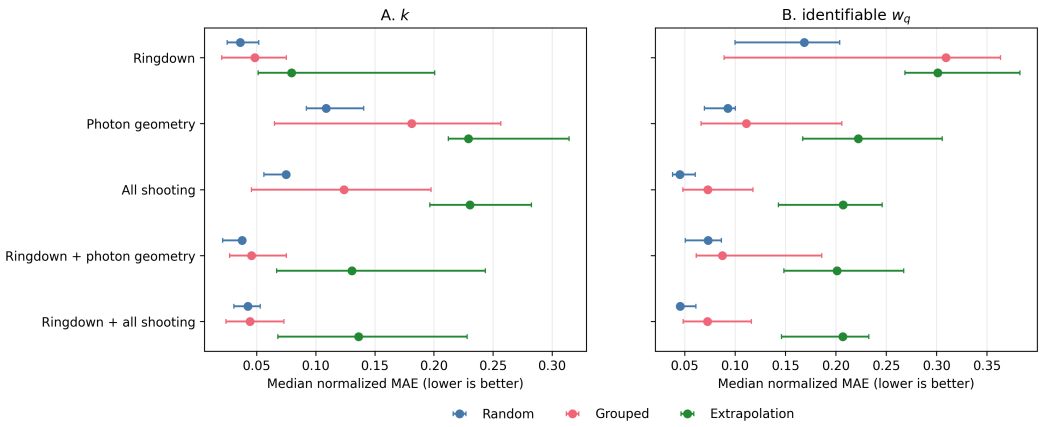


Figure 5: NMAE for the five primary feature sets under random, grouped physical interpolation, and directional extrapolation. Each dot preserves the predefined table statistic: the median of stored summary-row medians. To expose lower-level dispersion rather than suppress it through that hierarchy, each horizontal bar spans the 25th–75th percentile of the underlying NMAE records across the applicable folds, seeds, models, and, for extrapolation, directions. Thus the point and bar intentionally summarize different aggregation levels; directions remain separate in the machine-readable source before aggregation.

6.1 Traditional inverse baselines

To test whether the photon-geometry benefit belongs to the physical forward map rather than to one ML architecture, we evaluate non-learned nearest-model and local Jacobian inversions under exactly the same fixed physical holdouts. The nearest physical model fits a standardization transform on the training observables only, computes every Euclidean distance to the training catalog explicitly, and returns the (k, w_q) pair of the nearest training system. The local Jacobian inverse selects its reference from the training catalog, constructs a training-only local linear forward map, and applies the Moore–Penrose pseudoinverse. Calibration and test systems participate in neither construction.

For the nearest baseline, identifiable-w_q NMAE decreases from 0.123810 to 0.086580 under random interpolation, from 0.220833 to 0.148148 under grouped interpolation, and from 0.254161 to 0.195026 under directional extrapolation when photon geometry is added to ringdown. These are reductions of 30.1%, 32.9%, and 23.3%, respectively. Observable complementarity is therefore a property of the physical forward map rather than merely an effect learned by a particular estimator; nearest-neighbor inversion is not asserted to be optimal.

The same non-learned check supports redshift complementarity without making it a headline result: grouped identifiable-w_q NMAE is 0.2208/0.1585/0.1481 for ringdown, ringdown plus redshift, and ringdown plus photon geometry, respectively; the corresponding directional values are 0.2542/0.1932/0.1950. Both additions improve on ringdown, and their directional nearest-model performance is essentially comparable, supporting genuine physical information beyond the ringdown channel.

For ringdown plus photon geometry, the local Jacobian median k/identifiable-w_q NMAE is 0.0147/0.0259, 0.0184/0.0454, and 0.0661/0.1172 for random, grouped, and directional protocols. However, 1315 of 9950 raw predictions lie outside the analysis domain and were not clipped. Thus strong local inversion performance coexists with poor global robustness; favorable median local-linear performance does not establish a reliable global inverse. The full method comparison is shown in Fig. 6, with detailed values and conditioning associations retained in Appendix G.

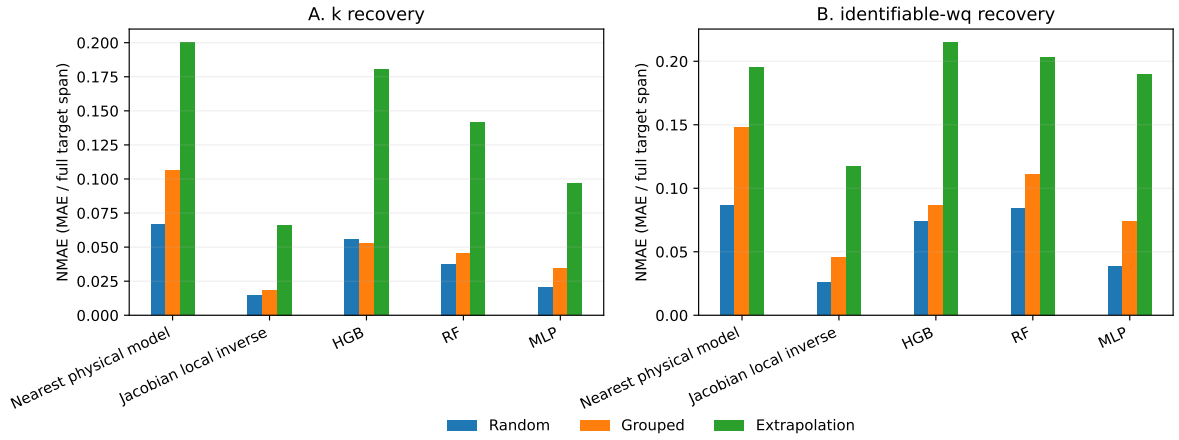


Figure 6: Traditional and learned inverse methods using ringdown plus photon geometry. Bars show median NMAE for k and identifiable w_q under random, grouped, and directional protocols. The nearest and Jacobian baselines are deterministic conditional on each predefined split, whereas archived ML summaries aggregate the existing folds and seeds. The Jacobian bars must be read with the unclipped out-of-domain diagnostic reported in the text.

Conditioning and coverage explain different aspects of error. In the error model, standardized training distance has estimate 0.330, extrapolation status 0.276, and log condition number 0.197. Physical conditioning generally agrees with grouped tree-based w_q performance, but it does not perfectly rank every model, target, or protocol. The MLP supplies a counterexample: it can have reasonable central metrics while producing catastrophic directional tails. The traditional-versus-learned comparison appears in Fig. 6; the conditioning associations are retained separately in Appendix G.

Nominal 90% conformal coverage is near its target under random interpolation (0.892/0.965 for k/identifiable w_q), but deteriorates under grouped (0.722/0.851) and directional evaluation (0.509/0.660). This degradation shows that calibration on one distribution does not ensure coverage after a physical holdout or directional shift. The detailed coverage distributions, evaluation-unit dependence, and $k = 0$ equivalence-class sensitivity are reported in Appendix F.5.

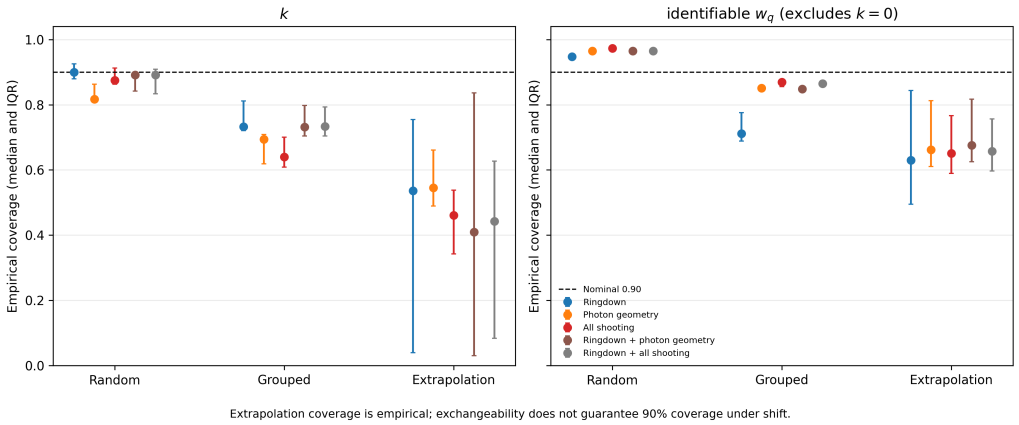


Figure 7: Empirical 90% split-conformal coverage by target and protocol. Ordinary w_q coverage excludes $k = 0$; extrapolation coverage is not guaranteed under distribution shift.

7 Numerical Convergence and Estimator Robustness

The physical forward features converge as the phase resolution increases, although the uniform 81-to-161-phase comparison leaves a small number of feature–point differences above the predefined tolerance. The targeted 161-to-321-phase audit resolves those differences on all 35 preselected systems: its median normalized feature change is 1.963×10^{-4}, and its 95th percentile is 7.234×10^{-3}. The ringdown features and Jacobian rank behavior remain unchanged. Figure 8 shows the reduction in feature changes. The selection protocol, acceptance thresholds, full-grid results, and ranking checks are documented in Appendix H.

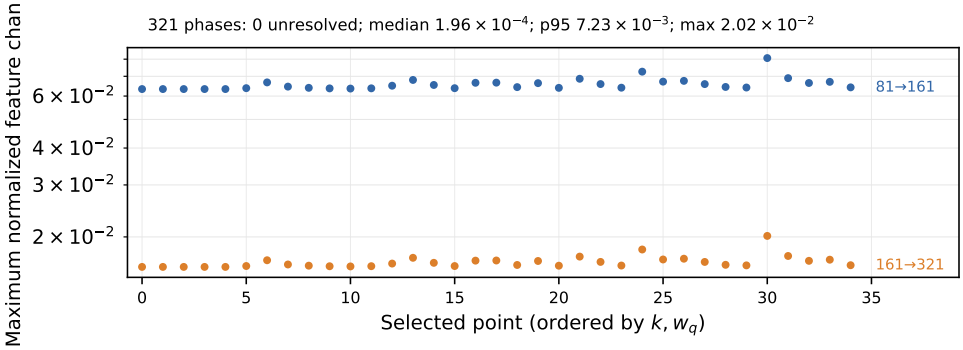


Figure 8: Targeted pointwise maximum normalized feature changes from 81 to 161 and 161 to 321 phases. The 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators.

Forward convergence does not imply inverse-estimator stability. We therefore evaluate the previously trained HGB, RF, and MLP estimators on refined features without retraining. The trees have unchanged median predictions but sensitive tails: the HGB and RF 95th-percentile normalized prediction shifts are 0.08122 and 0.08509, exceeding their prespecified stability tolerance. The MLP's catastrophic directional predictions largely predate the higher-resolution substitution. Thus the targeted forward audit passes, but the frozen estimators do not satisfy every robustness criterion. Figure 9 contrasts these failure mechanisms; full thresholds and counts are given in Appendix H.

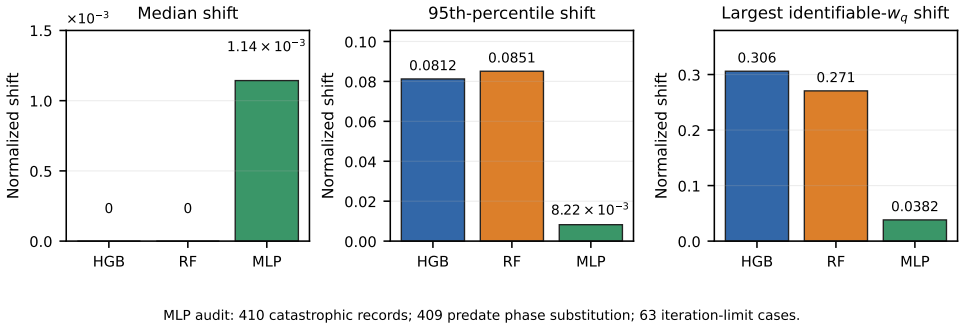


Figure 9: Frozen-estimator sensitivity after 161-to-321 forward convergence. Bars show the median, 95th-percentile, and largest identifiable-w_q normalized prediction shift for HGB, RF, and MLP. The annotation summarizes the corresponding MLP failure diagnostics. No predictions are clipped.

8 Discussion

The central lesson is that inverse-model design should begin with the information carried by the observables, not with a regression score. In this Kiselev test, leading-eikonal ringdown leaves a weak parameter direction; photon geometry and redshift add complementary information that makes identifiable w_q substantially easier to recover under grouped physical holdouts. The gain is also found with a non-learned nearest-model inverse, so it cannot be attributed solely to the trained architectures.

The exact $k = 0$ boundary provides an unusually clear control: w_q becomes physically indistinguishable there, independently of the inverse estimator. At finite k, the Jacobian diagnoses local parameter separation, but its conditioning does not reliably rank every prediction error. Training support, model architecture, and distribution shift matter in addition to local rank. The finite-grid collision audit finds no unresolved distant finite-k collision at its tested resolution, without establishing continuous global injectivity (Appendix I).

The observation-facing sensitivity analysis tests whether useful inverse recovery requires access to latent geometric descriptors. It does not. Restricting ringdown to Ω and λ and the sky channel to signed α and β weakens ringdown-only identifiability, but adding redshift or signed sky coordinates still improves the weakest parameter direction. These channels have different routes toward observational use: redshift is spectroscopic, whereas the sky coordinates here describe idealized rays. Definitions, normalization checks, and the numerical comparison are given in Appendix D.

The numerical audit identifies a second distinction. The refined forward calculation satisfies the targeted convergence criteria, yet frozen tree estimators retain split-crossing tail sensitivity and the MLP exhibits unstable directional extrapolation. Better forward resolution alone therefore does not repair an inverse estimator's response to off-distribution inputs [15, 13, 19].

These findings suggest an order of operations for future spectroscopy studies: first test parameter rank and degeneracies, then choose complementary observable channels, and only afterward optimize an inverse estimator under physically grouped and shifted evaluation. Extending this sequence to finite-multipole waveforms, realistic imaging or astrometry, spectroscopic likelihoods, and detector noise would establish how much of the theoretical complementarity remains operationally accessible. The limitations of the present forward and observation models are specified below.

9 Limitations

The ringdown model is leading-eikonal rather than a dedicated finite-ℓ perturbative spectrum. The study covers one static spherical Kiselev family. Shooting includes only the direct photon branch, a point emitter and point observer, and no radiative transfer, detector response, secondary images, strain likelihood, calibration error, observational measurement-noise model, or model discrepancy; the reported feature-noise tests are synthetic perturbations. The physical grid is finite. Historical proxy results remain synthetic. Although the targeted audit resolves higher-harmonic forward differences on its selected set, it is not a uniform 321-phase grid and tree tail sensitivity persists, and MLP extrapolation is unstable. These limitations preclude interpreting the results as an observational constraint. The finite-domain ambiguity audit does not prove continuous global injectivity; the 121-point grid can miss isolated or continuous degeneracies between sampled parameter values. Conformal coverage is sensitive to the representation of the exact $k = 0$ equivalence class, and the collapsed sensitivity design does not separate reduced Schwarzschild multiplicity from removal of non-identifiable nominal w_q labels. The present photon-geometry variables are idealized ray-level quantities; calling the restricted subset observation-facing does not make it detector-level or observational. A natural next step is to replace them with detector-level imaging or astrometric observables and realistic likelihoods, while retaining redshift as a directly interpretable spectroscopic channel.

10 Conclusion

Physical complementarity is supported: ringdown constrains k, while photon geometry supplies an independent direction needed for much of the w_q recovery. Importantly, this conclusion does not require the full latent geometric feature set. The restricted observation-facing analysis shows that complementary information and useful inverse recovery persist using only leading-eikonal Ω and λ, redshift, and signed sky coordinates. Observable complementarity also persists under a non-learned inverse baseline, while the local Jacobian inverse demonstrates that strong local recoverability can coexist with globally unstable predictions in this Kiselev benchmark. Random interpolation overstates recoverability relative to grouped and directional validation, and shifted conformal coverage is degraded and heterogeneous. The exact $k = 0$ boundary is structurally non-identifiable, whereas small nonzero k can be full-rank but only weakly identifiable because of practical ill-conditioning. Finally, numerical and distribution-shift audits are necessary before confident inverse claims: the forward features can converge while fitted estimators remain fragile. These conclusions hold within the predefined finite Kiselev grid and tested normalizations; they do not establish observational constraints, continuous global injectivity, generic beyond-Kerr validity, a uniform 321-phase result over all 121 systems, or universal scientific-ML guarantees. They motivate robust inverse methods and broader physical modeling before observational use.

Data Availability

The analysis code, machine-readable derived results included in the repository, fixed split assignments, and reproducibility audits are available in the project repository and are archived as version v1.0.0 on Zenodo [24]. This work uses simulated theoretical data and no observational data. A standard repository checkout can reproduce the headline aggregations from the included prediction tables without rerunning the physical grid. Full bitwise replay of the frozen-estimator audit additionally requires 5,520 serialized estimators (approximately 822 MB), which are not included in the repository

and do not yet have a permanent public archive; they are available from the author upon reasonable request. The repository includes the observation-facing sensitivity tables, model-specific metrics, figure source data, analysis script, and input-file SHA256 hashes.

A Historical Synthetic and Proxy Benchmark

The original dense benchmark used leading-eikonal curves, training-fold PCA, and synthetic geodesic proxies. Waveform-only RF achieved $R^2(k) = 0.8960$ and $R^2(w_q) = 0.9587$; HGB with proxy augmentation achieved 0.9793 and 0.9891. The rank-aware classifier reached macro F1 0.9928 with a 57.6% weak-identifiability fraction. PCA compressed correlated information; it did not create a new physical response direction. These historical results motivated the physical shooting study and are not substituted for it. Figure 11 serves a distinct role from the historical lens in Fig. 10: it compares a learned rank-aware class with the analytic weak-identifiability boundary rather than displaying conditioning or proxy regression-error changes.

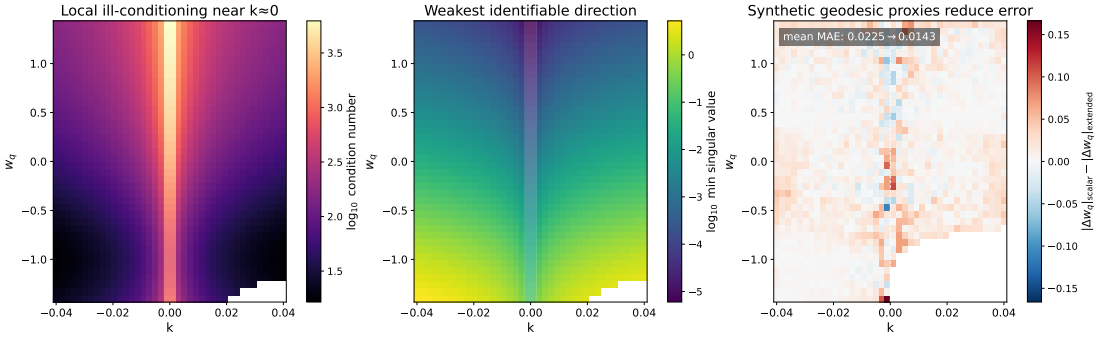


Figure 10: Historical analytic identifiability lens. Left: local Jacobian condition number, which grows near $k = 0$. Center: minimum singular value, showing the weakest identifiable direction. Right: change in identifiable-w_q error after adding historical synthetic geodesic proxies; the mean MAE changes from 0.0225 to 0.0143. White regions are invalid or missing source-grid points and are not interpolated. The third panel is not physical ray tracing, and proxy augmentation does not resolve exact $k = 0$ rank loss.

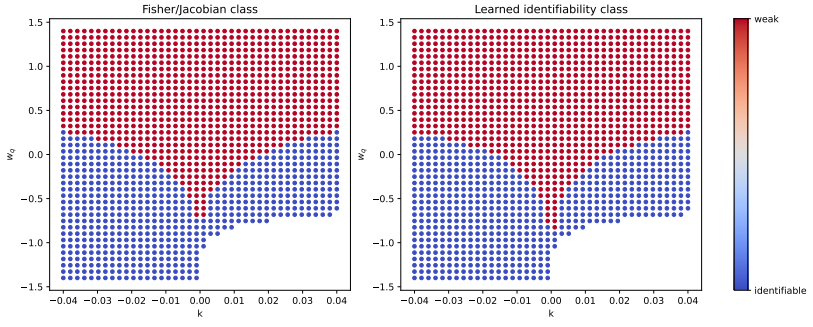


Figure 11: Historical dense synthetic rank-aware identifiability boundary. Exact $k = 0$ rank loss is structural, whereas the finite-width weak-identifiability region depends on precision. This historical benchmark motivated, but does not replace, the validated physical shooting analysis.

B Full Split Definitions

The assignment table contains 115 predefined split specifications over five seeds. For each seed it contains one random interpolation split, 18 grouped splits (nine held-out blocks for each of two 3×3 layouts), and four directional splits. Random splitting holds out 20% of the 121 physical systems for test and assigns 20% of the remainder to calibration. The grouped layouts use index shifts $(0, 0)$ and $(1, 1)$ before forming the nine contiguous blocks; each block is held out once and 20% of the remaining systems calibrate the fit. Directional tests separately hold out the highest three k levels, lowest three k levels, lowest three w_q levels, and highest three w_q levels, with 20% of the complementary pool used for calibration. Training, calibration, and test systems are disjoint. All $k = 0$ systems remain included; they contribute to k scoring but not ordinary w_q scoring because w_q is unidentifiable there. Every scaler, noise scale, feature transform, and model is fitted on training data only. The same predefined split assignments are used throughout.

C Feature Definitions

Table 2 lists the predefined feature families. Except for the two orbital invariants and four ringdown scalars, each phase-resolved quantity is represented by its mean, amplitude, reference value where defined, and sine/cosine coefficients through the third harmonic.

Table 2: Predefined physical feature families and exact dimensions.

Family	Physical quantities	Source	Dim.
Ringdown	Δr, photon-sphere radius, Ω, Lyapunov exponent	static metric/eikonal model	4
Orbital	radial azimuthal period, apsidal advance; summaries of r and p_r	timelike emitter	20
Photon geometry	summaries of impact parameter and signed sky coordinates	direct null rays	27
Redshift	summaries of photon redshift	emitter/ray/observer	9
Timing	summaries of excess delay and relative arrival time	direct null rays	17
All shooting	orbital, photon geometry, redshift, and timing	physical shooting	73
Ringdown + photon geometry	union of the corresponding families	combined	31
Ringdown + all shooting	union of the corresponding families	combined	77

D Observation-Facing Feature Sensitivity

This secondary analysis tests whether the complementarity conclusion depends on latent theoretical or ray-geometric descriptors. The restricted ringdown set, denoted $\mathrm{RD}_{\mathrm{obs}}$, contains only Ω and λ; it excludes Δr and r_{ph}. The redshift addition contains all nine existing phase summaries of z. The sky addition contains the nine summaries of signed α and the nine summaries of signed β, while excluding all impact- parameter summaries. We also examine the combined redshift-plus-sky set. Orbital and timing features are excluded throughout. We use "observation-facing" only to distinguish these quantities from latent geometric descriptors: Ω and λ remain leading-eikonal quantities, while α and β are idealized ray-level sky coordinates rather than detector-level measurements.

The two numerical layers follow the same protocol as the primary analysis. The Jacobian columns in Table 3 use the nominal 81-phase feature set, q95–q05 feature scaling, unchanged parameter spans and finite-difference stencils, and analytic $k = 0$ null handling. The inverse columns use the separately validated 161-phase representation and the same 115 predefined train/calibration/test splits, seeds, training-only preprocessing, model hyperparameters, and identifiable-w_q scoring rule.

Accordingly, the Jacobian and inverse-learning columns refer to their respective validated resolutions rather than to a single common-resolution feature table.

Table 3: Observation-facing sensitivity summary. Jacobian medians use the nominal 81-phase archive; inverse NMAE uses the validated 161-phase representation. "Near." is the non-learned nearest-physical-model baseline. All w_q scores exclude $k = 0$.

Feature set	Dim.	med. $\sigma_{\min}$	med. $\kappa(J)$	Grouped	Directional	Near. grp.	Near. dir.
$\mathrm{RD}_{\mathrm{obs}}$	2	0.0097	136.69	0.3886	0.4239	0.4238	0.4377
+ redshift	11	0.6215	4.47	0.0985	0.1750	0.1417	0.1932
+ signed sky	20	0.6999	4.89	0.0904	0.2179	0.1308	0.1950
+ redshift + sky	29	0.7310	5.59	0.0807	0.1949	0.1190	0.1934

Redshift and sky improve $\sigma_{\min}$ at every eligible finite-k grid point under the nominal scaling. Their median pointwise $\sigma_{\min}$ gains relative to $\mathrm{RD}_{\mathrm{obs}}$ are $56.80\times$ (paired-bootstrap 95% interval $[43.99, 64.10]$) and $72.11\times$ ($[60.54, 83.60]$), respectively; these ratios are secondary to the absolute values in Table 3. The ordering is qualitatively stable under IQR and population-standard-deviation scaling, as shown in Table 4. Feature-family balancing is not reported because partially selecting the ringdown and photon families makes the earlier whole-family weighting convention ambiguous.

Table 4: Scaling sensitivity of median pointwise gains relative to $\mathrm{RD}_{\mathrm{obs}}$. Each entry is $\sigma_{\min}$ gain / conditioning improvement.

Scaling	+ redshift	+ signed sky
q95–q05	56.80 / 33.10	72.11 / 25.47
IQR	81.85 / 37.63	96.42 / 26.66
Population standard deviation	51.50 / 34.09	62.49 / 32.27

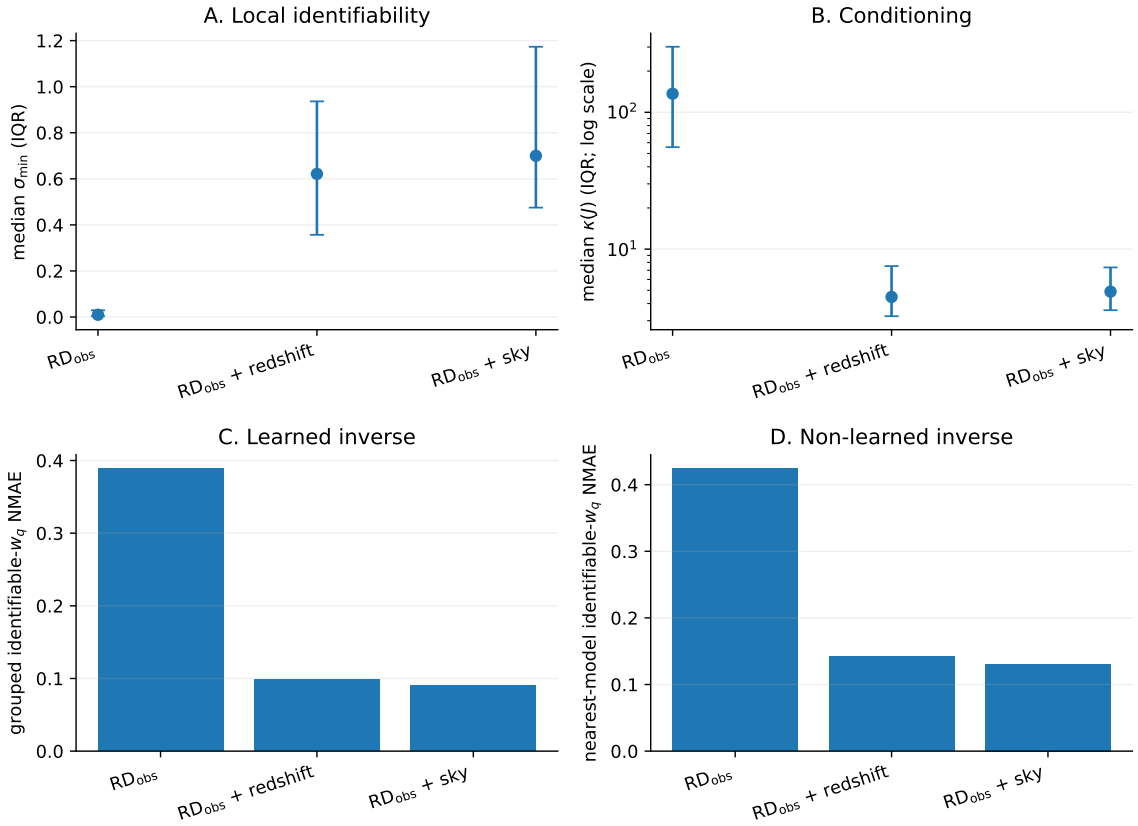


Figure 12: Observation-facing sensitivity. Panels A and B show nominal 81-phase finite-k medians and interquartile ranges; the conditioning axis is logarithmic to show the large baseline and the two restricted additions without compressing the latter. Panels C and D show grouped identifiable-w_q NMAE from the 161-phase learned-model aggregate and non-learned nearest-model baseline. Higher $\sigma_{\min}$ and lower $\kappa(J)$/NMAE are favorable.

The learned-model medians retain the improvement for each architecture, as shown in Table 5.

Table 5: Model-specific identifiable-w_q NMAE for the three primary restricted comparisons.

Protocol	Model	RD_{obs}	+ redshift	+ signed sky
Grouped	HGB	0.3886	0.0985	0.0904
	RF	0.4592	0.1011	0.1108
	MLP	0.3862	0.0714	0.0710
Directional	HGB	0.3935	0.1983	0.2162
	RF	0.4236	0.2023	0.2040
	MLP	0.4361	0.1655	0.2613

The combined redshift-plus-sky set has the best grouped aggregate NMAE (0.0807) but not the best directional aggregate (0.1949 versus 0.1750 for redshift), reinforcing that more features do not universally improve extrapolation.

This sensitivity check does not implement radiative transfer, image reconstruction, a point-spread function, a detector likelihood, or an observational noise model. It therefore supports robustness of the theoretical complementarity conclusion, not an observational constraint. Complete model-

specific, nearest-model, pointwise-Jacobian, scaling, and provenance tables are provided with the repository.

E Pointwise Finite-Grid Identifiability Maps

The grid-wide medians and interquartile ranges in the main text summarize the overall observable-family ordering. Figure 13 retains the corresponding pointwise structure: it shows where local identifiability is strong or weak within the analysis domain without interpolating missing or invalid systems.

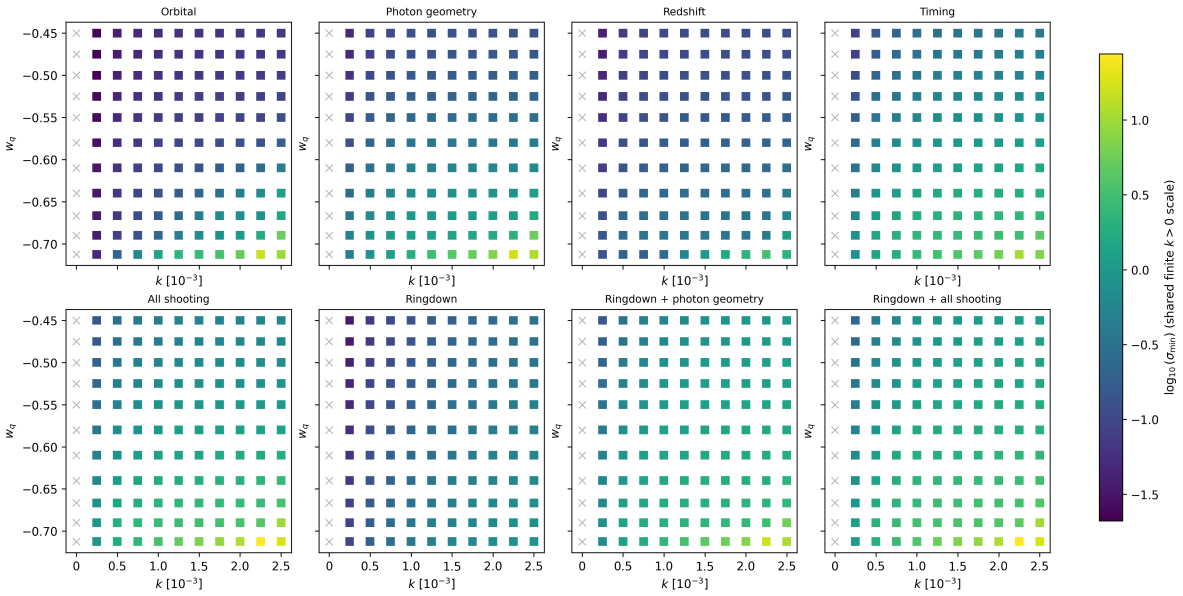


Figure 13: Grid maps of $\log_{10} \sigma_{\min}$. Crosses at $k = 0$ mark exact structural rank loss rather than missing simulations. Missing or invalid values are not interpolated, and all panels use one finite-k color scale.

F Additional Inverse-Learning Diagnostics

F.1 Additional interpretation and sensitivity checks

To distinguish improved physical observability from architecture-specific learning effects, we also evaluate non-learned nearest-model and local-linear inverse baselines under the same physical holdout protocols. The nearest-model improvement confirms an estimator-independent geometric contribution. Changes in local conditioning, however, do not consistently predict pointwise recovery across all models and protocols: Spearman associations have mixed signs and broad bootstrap intervals. Jacobian conditioning diagnoses local physical distinguishability, but it is not a complete predictor of finite-data error or behavior under distribution shift, and no causal relationship is inferred. The redshift comparison sharpens this distinction: redshift is more directly connected to a measurable spectroscopic quantity than the idealized ray-level photon-geometry variables used here, but the present synthetic benchmark has no detector likelihood, radiative-transfer model, or observational noise model.

As a secondary observation-facing sensitivity analysis, we repeated the comparison after restricting ringdown to the leading-eikonal quantities Ω and λ, excluding Δr and $r_{\rm ph}$, and restricting the

sky channel to signed α and β, excluding impact parameter b. Ringdown-only local identifiability is substantially weaker under this restriction: the finite-k grid medians are $\sigma_{\min} = 0.0097$ and $\kappa(J) = 136.69$. Nevertheless, adding redshift or the signed sky coordinates raises the median $\sigma_{\min}$ to 0.6215 or 0.6999 and lowers the median $\kappa(J)$ to 4.47 or 4.89. Grouped and directional inverse tests, together with the non-learned nearest-model baseline, independently support the same qualitative conclusion: the latent photon-orbit quantities substantially strengthen ringdown-only identifiability, whereas the qualitative redshift and sky-coordinate complementarity persists after they are removed. These features are observation-facing, not detector-level; α and β remain idealized ray-level observer-sky coordinates. Appendix D gives the definitions, resolution provenance, scaling checks, and complete secondary results.

Jacobian conditioning answers whether nearby parameter directions are separated by the chosen observables. Training distance answers whether a fitted estimator has support where it is evaluated. Neither statistic subsumes the other. Consequently, $\sigma_{\min}$ can predict broad observable-set ordering without ranking every empirical model or protocol.

The Jacobian SVD magnitudes also depend on the declared observable and parameter normalizations. The nominal q95–q05/full-span convention remains the primary analysis; IQR, population-standard-deviation, and feature-count-balanced checks preserve the median ringdown-plus-photon-geometry ordering but alter its magnitude. Pointwise conditioning is spatially nonuniform under every scheme, so this sensitivity check supports qualitative complementarity rather than a scale-invariant gain factor.

Tree ensembles are piecewise and can show zero median response followed by large split-crossing tails. The MLP is smoother under the targeted feature substitution but extrapolates to physically unreasonable values in archived directional tests. These mechanisms are estimator-specific manifestations of a general scientific-ML risk: a converged forward simulator does not ensure a stable inverse surrogate [15, 13, 19].

What is specific to Kiselev is the metric deformation and exact $k = 0$ loss of w_q. Methodologically, this boundary is useful because the correct identifiability answer is known analytically before any inverse model is fitted: the pipeline must return no physical w_q information at $k = 0$. Recovering that ground-truth null control checks the diagnostic in this controlled benchmark, but does not establish universal validity. What may generalize is the workflow: establish forward rank, seek complementary observables, partition by physical systems, test directional shift, and perturb frozen estimators at independently measured numerical scales.

The Jacobian analysis is local by construction. The finite-grid collision audit provides a complementary nonlocal check across the sampled parameter domain: no unresolved distant finite-k collision is found at the tested resolution, although photon geometry rearranges local neighbor ordering. Together with the improved nearest-model recovery, this shows that improved distant sampled separation need not imply preservation of every local observable-space neighborhood.

F.2 Additional model-level results

HGB and RF reproduce the central grouped photon-geometry improvement for identifiable w_q. The MLP is retained as a stress test rather than omitted: its catastrophic errors are unclipped and included in aggregate summaries.

F.3 Four directional extrapolation tests

Directions are stored independently in the prediction tables before any median summary. High-to-low and low-to-high tests differ in training support and cannot be interpreted as interchangeable folds.

F.4 Conformal calibration and rejection

Split-conformal residual quantiles are fitted on calibration subsets. The exchangeability guarantee applies to matching distributions; shifted coverage is reported empirically. Rejection curves rank uncertainty but do not repair distribution shift.

F.5 Finite-grid and coverage uncertainty

Nominal 90% conformal coverage is 0.892/0.965 for random k/w_q, but falls to 0.722/0.851 under grouped interpolation and 0.509/0.660 under extrapolation. Each value is the median of archived model–feature-set coverage rows (15 rows for random and grouped protocols; 60 direction–model–feature-set rows for extrapolation), after each row pools its fixed fold/seed prediction indicators. The corresponding row-level IQRs are 0.833–0.908/0.951–0.973, 0.676–0.747/0.840–0.863, and 0.311–0.689/0.585–0.844. Because models, features, seeds, and layouts reuse physical systems, these rows are not independent Bernoulli trials and no single pooled binomial interval is assigned to their median. Wilson intervals for each individual evaluation unit and all four directional summaries are retained in Appendix F.5. Shifted coverage remains empirical, not a retained finite-sample guarantee. Figure 7 separates coverage by target and protocol.

In the collapsed-equivalence check, the 11 identical Schwarzschild rows are replaced by one class without a nominal w_q target, while all finite-k assignments remain unchanged. Across learned model–feature summaries, random, grouped, and directional k NMAE changes from 0.0471/0.0528/0.1714 to 0.0387/0.0594/0.1567; identifiable-w_q NMAE changes only from 0.0621/0.0875/0.2038 to 0.0623/0.0859/0.2032. Coverage is more sensitive: identifiable-w_q values become 0.929/0.763/0.509. Thus the primary qualitative recovery and complementarity conclusions are stable under this sensitivity analysis, while conformal coverage is more sensitive to the representation of the $k = 0$ equivalence class. The collapsed analysis is treated as a sensitivity analysis rather than as a replacement for the primary analysis.

Table 6 reports uncertainty summaries for the finite analysis grid. The Jacobian percentile intervals use 10,000 paired physical-system bootstrap resamples (seed 20261002). Coverage entries are the median [IQR] across the stored rows that define the manuscript headline; the evaluation-unit count is given in parentheses. At the lower level, each unit is a single fold/seed/model/feature/target evaluation on distinct held-out systems. Its covered numerator, eligible denominator, and 95% Wilson interval are stored in machine-readable form. Units overlap across models, features, seeds, and layouts, so their Wilson intervals are diagnostics rather than ingredients of a falsely pooled confidence interval.

Table 6: Finite-grid stability and empirical-coverage dispersion. Coverage is shown for k/identifiable w_q; $k = 0$ remains in k scoring and is excluded from ordinary w_q scoring.

Summary	k	identifiable w_q
Random coverage	0.892 [0.833, 0.908] (75)	0.965 [0.951, 0.973] (75)
Grouped coverage	0.722 [0.676, 0.747] (1350)	0.851 [0.840, 0.863] (1350)
Directional aggregate	0.509 [0.311, 0.689] (300)	0.660 [0.585, 0.844] (300)
High k to low k	0.315 [0.042, 0.600] (75)	0.582 [0.509, 0.659] (75)
Low k to high k	0.297 [0.012, 0.448] (75)	0.939 [0.894, 0.973] (75)
High w_q to low w_q	0.709 [0.515, 0.891] (75)	0.647 [0.580, 0.710] (75)
Low w_q to high w_q	0.564 [0.461, 0.742] (75)	0.627 [0.587, 0.680] (75)
Jacobian median gain	$\sigma_{\min}$: 4.5429 [4.2840, 4.9440]; $\kappa(J)$: 2.2819 [1.8382, 2.6254] ($n = 110$)	

F.6 Equivalent Schwarzschild-record sensitivity

The 11 nominal $k = 0$ rows have different nominal w_q labels but identical values for all 77 predefined physical features. Their maximum absolute and q95–q05-standardized feature differences are both

exactly zero in the feature table. Because the predefined splits treat nominal rows separately, physically equivalent Schwarzschild records cross train/test in 45 of 115 split specifications and train/calibration in 96. This is nominal-record separation but not physical-equivalence separation.

As a secondary check, the 11 rows are collapsed to one equivalence record whose partition is the majority nominal $k = 0$ role in each split; ties prefer training, then calibration, then test. This minimizes assignment changes without selecting a nominal w_q. All 110 finite-k assignments remain unchanged, and the collapsed record is excluded from w_q fitting, calibration, testing, and scoring. Table 7 compares the nominal and collapsed medians across learned model–feature protocol summaries. Secondary models use the predefined classes, hyperparameters, and seeds with training-only preprocessing.

Table 7: Nominal (N) versus collapsed-equivalence (C) sensitivity. Each cell is N $\rightarrow$ C; ordinary w_q quantities use finite k only.

Protocol	k NMAE	identifiable w_q NMAE	k coverage	identifiable w_q coverage
Random	0.0471 $\rightarrow$ 0.0387	0.0621 $\rightarrow$ 0.0623	0.892 $\rightarrow$ 0.903	0.965 $\rightarrow$ 0.929
Grouped	0.0528 $\rightarrow$ 0.0594	0.0875 $\rightarrow$ 0.0859	0.722 $\rightarrow$ 0.725	0.851 $\rightarrow$ 0.763
Directional	0.1714 $\rightarrow$ 0.1567	0.2038 $\rightarrow$ 0.2032	0.509 $\rightarrow$ 0.479	0.660 $\rightarrow$ 0.509

In the nominal predictions, 147 distinct $k = 0$ split/test-system cases have an equivalent Schwarzschild training record and 55 do not. Their mean k NMAE is respectively 0.0355 and 0.2792, with empirical coverage 0.990 and 0.160; equivalent training support therefore makes this boundary test locally easier. Nevertheless, the collapsed aggregate changes have mixed signs rather than a uniform optimistic bias. Random interpolation remains easier than grouped and directional testing. HGB, RF, and the nearest-physical-model check retain a positive identifiable-w_q reduction from photon geometry in every protocol; the MLP remains the documented counterexample. Shifted conformal coverage stays degraded and heterogeneous, and the frozen-estimator robustness audit is unchanged. This majority-role collapse is a least-change sensitivity design, not a unique equivalence-aware resampling protocol. It changes both the multiplicity of the Schwarzschild equivalence class and the removal of non-identifiable nominal w_q labels from fitting and calibration, so these two effects are not separately identified.

F.7 Noise and learning curves

Grouped one-percent feature noise yields ringdown-plus-photon-geometry NMAE 0.0699/0.0853 for k/w_q. Learning curves are still improving at 78 training points, so data limitation remains part of the interpretation.

G Traditional Inverse Baselines

Table 8 gives the complete ringdown-plus-photon-geometry comparison. Each entry is median k/identifiable-w_q NMAE. The nearest and local-linear methods are deterministic for a fixed predefined split; their rows are summarized over the same split assignments, while archived learned model rows retain the existing fold/seed aggregation.

Table 8: Traditional and learned inversion with ringdown plus photon geometry. NMAE is MAE divided by the full target span; $k = 0$ remains in k scoring but is excluded from identifiable-w_q scoring.

Method	Random	Grouped	Directional
Nearest physical model	0.0667/0.0866	0.1063/0.1481	0.2000/0.1950
Local Jacobian inverse	0.0147/0.0259	0.0184/0.0454	0.0661/0.1172
HGB	0.0560/0.0740	0.0530/0.0865	0.1803/0.2146
RF	0.0378/0.0844	0.0457/0.1113	0.1418/0.2033
MLP	0.0206/0.0383	0.0344/0.0738	0.0966/0.1896

The primary Jacobian predictions are raw and unclipped: 1315 of 9950 fall outside $0 \leq k \leq 0.0025$ or $-0.7125 \leq w_q \leq -0.45$. Reference condition number, reference $\sigma_{\min}$, and local observable-space distance are stored for every prediction. Figure 14 reports the supported pointwise/fold-level conditioning analysis. Most bootstrap intervals span zero, and the significant nearest-model associations are negative; these mixed results reinforce that local conditioning is not a complete estimator-error model.

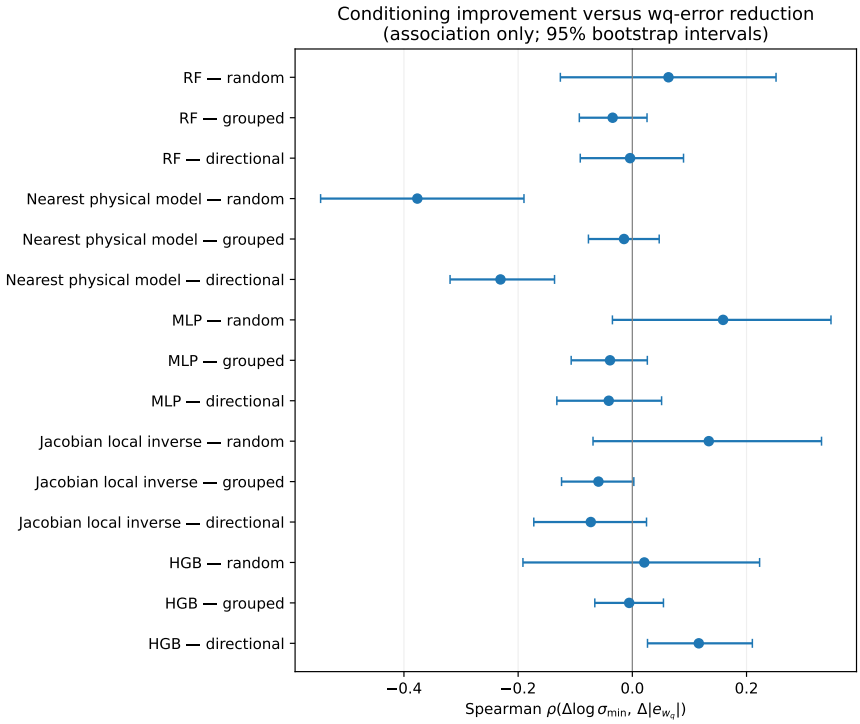


Figure 14: Spearman association between the change in log minimum singular value and the reduction in absolute identifiable-w_q error when photon geometry is added to ringdown. Points show method/protocol estimates and bars are 95% bootstrap intervals. Associations are descriptive and do not imply causation.

H Numerical Validation and Reproducibility

H.1 Numerical-validation thresholds

We first ask whether the physical forward calculation converges as phase resolution increases. Figure 8 addresses that question. For feature i at resolutions N_1 and N_2, the frozen comparison is

$$\delta_i = \frac{|F_i^{(N_2)} - F_i^{(N_1)}|}{\max(s_i, 10^{-12})}, \tag{6}$$

where s_i is the global q95–q05 scale computed on the nominal 81-phase training domain. The floor prevents division by numerical zero and is not fitted to either higher-resolution result. A feature–point comparison is classified as converged for $\delta_i \leq 0.01$, marginal for $0.01 < \delta_i \leq 0.05$, and unresolved for $\delta_i > 0.05$; non-finite, missing, or schema-mismatched values are also unresolved by protocol. Here "unresolved" denotes failure of this numerical-resolution criterion, which was fixed before evaluation, rather than a failed physical shooting phase.

The uniform 161-phase study completes all 121 physical points. From 9317 feature–point comparisons, 240 exceed the 0.05 normalized-change threshold and are therefore unresolved under the frozen criterion; no failed physical point is implied. The median normalized feature change is 7.908×10^{-4} and the maximum is 0.08069. Across frozen scored predictions, the median normalized shift is 9.482×10^{-4} and the 95th percentile is 0.02606. The median aggregate NMAE change is 6.880×10^{-4}, but the maximum is 16.8497 because of unclipped MLP directional extrapolation outliers.

The targeted 321-phase audit selects 35 points deterministically before reading their 321-phase outcomes. The selection is the deduplicated union of pre-321 signals: the four largest 81-to-161 shifts in each of the feature, HGB, and RF rankings; the eight largest MLP shifts; fixed boundary and center points; four directional representatives; three points at each conditioning extreme; and one representative from each of nine grouped blocks. Its forward acceptance conditions, recorded before any 321-phase trajectory was evaluated, require median normalized change below 0.001, 95th-percentile change below 0.01, unchanged ringdown, no systematic family-wide trend, and unchanged Jacobian rank behavior. All selected points complete. Median and 95th-percentile normalized 161-to-321 feature changes are 1.963×10^{-4} and 7.234×10^{-3}; no comparison remains unresolved, ringdown is exactly unchanged, no systematic family-wide trend is found, and Jacobian rank is preserved. The targeted forward audit therefore passes the criteria specified in advance. This is not a uniform 321-phase proof over the full 121-point grid.

Within the selected 35-system set, the earlier resolution ranking was strongly preserved: the point-level maximum changes had Spearman $\rho = 1.000$, while the point-level 95th-percentile changes had $\rho = 0.997$. All top-five and top-ten systems were retained under the later maximum-change ranking. Because 321-phase values exist only for these 35 systems, this does not establish that they contain the largest changes over the full grid.

We next ask whether the previously trained estimators remain stable after the forward map has converged. Figure 9 applies those frozen estimators to the refined features; no model is retrained.

Forward acceptance and inverse-estimator acceptance are separate. For HGB and RF individually, the prespecified inverse criteria require median normalized prediction shift below 0.005, 95th-percentile shift below 0.03, maximum identifiable-w_q shift below 0.1, aggregate NMAE change below 0.02, and unchanged grouped complementarity. The written protocol permits an identifiable-w_q exceedance of the 0.1 scale only when it is traced to an unresolved feature; because no targeted feature comparison remains unresolved, that caveat does not relax the criterion here.

Frozen trees remain tail-sensitive: HGB and RF 95th-percentile normalized shifts are 0.08122 and 0.08509, and their largest identifiable-w_q shifts are 0.30593 and 0.27053. Their median shifts are zero because many tree predictions do not cross a split threshold. The fixed perturbation ensembles and fixed Extra Trees smoother do not pass all stability and grouped-performance criteria. Of 410 catastrophic MLP records, 409 are already catastrophic at 161 phases, none is materially worsened by the 321-phase substitution, 63 reach the iteration limit, and no numerical overflow is detected. Thus, the forward map satisfies the convergence criterion, whereas the frozen estimators do not satisfy all robustness criteria.

Typical predictions are stable, but the tails are not. The earlier MLP maxima are not representative of every estimator, while the targeted tree failures show that the remaining issue also cannot be attributed solely to the MLP. Figure 9 separates the estimator mechanisms. Tree medians remain exactly unchanged because most perturbations do not cross a split, yet their tails do. The MLP reacts more smoothly to phase substitution, while its catastrophic directional records overwhelmingly predate that substitution.

The frozen resolution protocol uses the nominal 81-phase q95–q05 feature scales and the 10^{-12} denominator floor defined in Section 7. Its feature-level boundaries are 0.01 (converged) and 0.05 (marginal); values above 0.05 are unresolved. Targeted forward acceptance uses median/p95 boundaries of 0.001/0.01 together with unchanged ringdown, no family-wide systematic trend, and unchanged rank behavior. Frozen-tree acceptance is assessed separately using median/p95 prediction-shift boundaries of 0.005/0.03, maximum identifiable-w_q shift 0.1 subject to the unresolved-feature caveat, aggregate NMAE change 0.02, and preserved grouped complementarity. All these criteria were recorded before evaluation of the 321-phase trajectories.

The 35 systems are the deduplicated union of ranked 161-phase stress signals, fixed boundary/center cases, directional representatives, conditioning extremes, and grouped-block representatives. Machine-readable point-selection metadata are provided with the repository. Point-level maximum and 95th-percentile 81-to-161 rankings are strongly preserved at 161-to-321 resolution ($\rho = 1.000$ and 0.9972, respectively), with top-five and top-ten maximum membership unchanged. This selected-set check does not rank the 86 systems for which no 321-phase calculation exists.

The 35-point 321-phase run takes 5318.2 s. Maximum hit, timelike-constraint, null-constraint, and impact-parameter drift errors are respectively 7.0913×10^{-8}, 3.1303×10^{-13}, 7.6214×10^{-12}, and 1.7932×10^{-11}. The Schwarzschild arrival-time comparison in Fig. 15 separates the visually overlapping phase-resolved series from their actual-scale residuals.

H.2 Additional complementarity audit details

The nominal 81-phase finite-grid bootstrap 95% intervals for the median pointwise photon-geometry improvements are $[4.28, 4.94]$ for $\sigma_{\min}$ and $[1.84, 2.63]$ for conditioning. They resample the 110 eligible finite-k systems and are not astrophysical-population confidence intervals. Ringdown plus all shooting has grid medians $(\sigma_{\min}, \kappa(J)) = (1.5312, 4.4974)$; the corresponding 161-phase results are $(1.5319, 4.4957)$, compared with $(1.2326, 4.0733)$ for ringdown plus photon geometry. The normalization stress tests yield median $\sigma_{\min}$/conditioning improvement factors of 5.88/1.87 (IQR), 3.76/2.40 (population standard deviation), and 2.95/2.22 (feature-family-count balancing). Every finite-k point has a $\sigma_{\min}$ gain above one; conditioning improves at 68.2–89.1% of points across these schemes, without a change in numerical rank. Numerical factors are therefore normalization-dependent even though the median improvement is qualitatively stable.

H.3 Jacobian normalization sensitivity

The nominal 81-phase Jacobian was recomputed from the corresponding feature grid using unchanged finite-difference stencils and parameter spans. Table 9 reports medians of pointwise finite-k ratios with their interquartile ranges; exact $k = 0$ rank-loss points are kept as a separate control and excluded from ratios involving zero $\sigma_{\min}$.

Table 9: Observable-normalization sensitivity. Fractions give finite-k points with improvement greater than one; "balanced" is the secondary $1/\sqrt{n_{\text{family}}}$ row-block stress test.

Scheme	$\sigma_{\min}$ gain, median [IQR]	gain > 1	κ improvement, median [IQR]	improve > 1
Nominal q95–q05	4.543 [3.845, 6.201]	100%	2.282 [1.207, 3.455]	80.0%
IQR	5.882 [4.415, 7.798]	100%	1.870 [0.590, 3.031]	68.2%
Population std.	3.763 [3.071, 5.223]	100%	2.401 [1.368, 2.954]	83.6%
Family-count balanced	2.954 [2.425, 3.942]	100%	2.225 [1.881, 2.755]	89.1%

All four schemes retain 110 eligible finite-k points, preserve all 11 exact $k = 0$ rank-one controls, and produce no finite-k numerical-rank change. Raw ringdown/combined grid medians of $(\sigma_{\min}, \kappa)$ are respectively $(0.2782, 6.6112)/(1.2326, 4.0731)$ for the nominal scheme, $(0.5553, 6.6688)/(2.7467, 5.0306)$ for IQR, $(0.8841, 6.6075)/(3.5168, 3.8107)$ for population standard deviation, and $(0.1391, 6.6112)/(0.4117, 3.3974)$ for the balanced stress test. No relevant feature scale required flooring or exclusion.

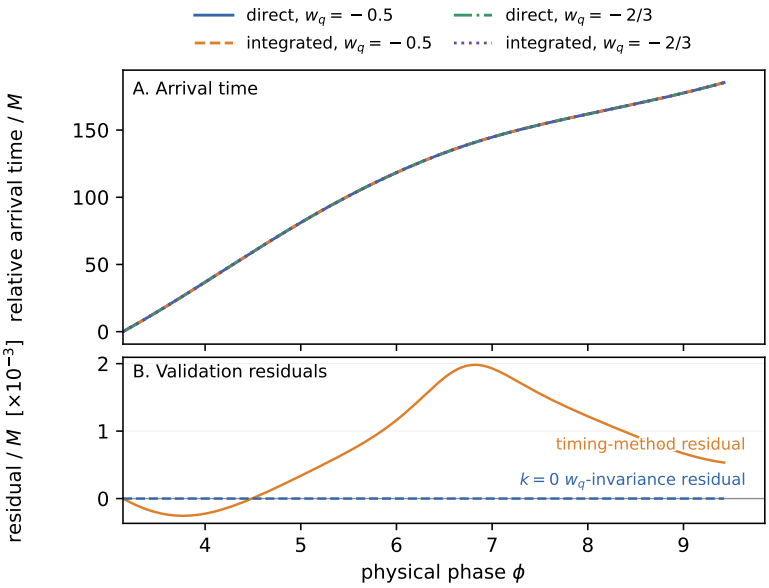


Figure 15: Schwarzschild arrival-time validation. The four phase-resolved series are the direct and redshift-integrated arrival-time constructions for each of two nominal w_q values. At $k = 0$ those values describe the same Schwarzschild geometry. In the lower panel, the orange residual checks timing consistency, $T_{\text{direct}} - T_z$, where $T_z = \int (1 + z)\, d\tau_{\text{emit}}$, and the blue residual checks exact $k = 0$ w_q invariance, $T_{\text{direct}}(w_q = -0.5) - T_{\text{direct}}(w_q = -2/3)$. Both are shown on their actual numerical scales. The independently integrated T_z is a validation quantity, not the timing feature used for ML inference.

For the small-deformation comparison, the full observer tracks nearly overlap on their physical scale. Figure 16 separates those tracks from the small but resolved matched-phase residuals.

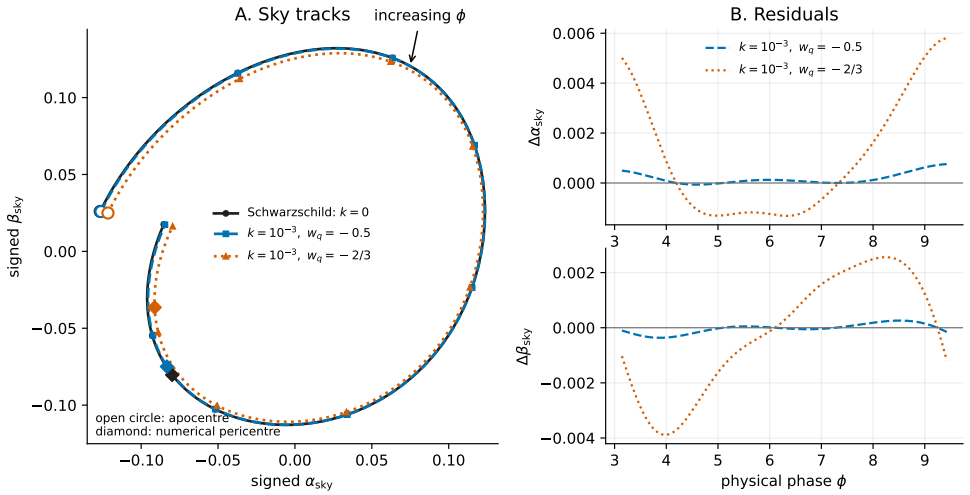


Figure 16: Static-observer-tetrad sky tracks and actual-scale residuals for the same $M = 1$, $r_p = 8M$, $r_a = 12M$, observer $(0, 0, -80M)$ direct-branch benchmark. Open circles mark $\phi = \pi$ apocentre, diamonds mark numerical pericentre, and the arrow gives increasing phase. Residuals are pointwise differences from Schwarzschild at exactly matched successful phases; they are neither normalized for visibility nor interpolated across failures.

Matched Schwarzschild and Kiselev direct rays are compared on their actual coordinate scale in Fig. 17.

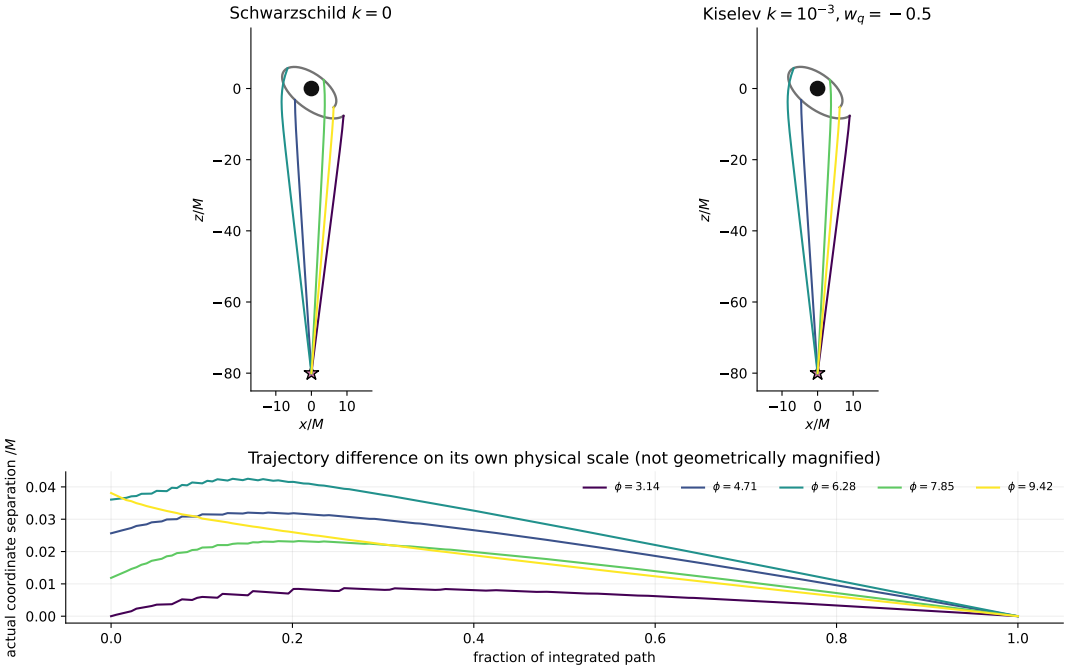


Figure 17: Matched-phase direct rays for Schwarzschild and $k = 10^{-3}$, $w_q = -0.5$ with $M = 1$, $r_p = 8M$, $r_a = 12M$, and observer $(0, 0, -80M)$. The lower panel gives physical coordinate separation versus integrated path fraction. Differences are shown on their actual scale and are not geometrically magnified.

H.4 Uniform 161-phase convergence audit

All 121 points complete in 4473.2 s with zero failed points. The audit preserves the exact $k = 0$ boundary and uses no mixed-resolution row, interpolation, or proxy substitution.

H.5 Reproducibility summary

The physical grid contains 121 points. The uniform 161-phase audit completed all points, and the targeted 321-phase audit completed all 35 preselected points. Exact $k = 0$ rank loss is preserved. The predefined split assignments are reused, preprocessing is fitted on training data only, and the traditional inverse numerical audit passed. The independent physics cross-check reported in Section 3 also passed. Additional implementation and reproducibility details are provided in the repository.

I Finite-Domain Observable-Ambiguity Audit

This audit evaluates all pairwise distances on the predefined 121-point physical grid. It cannot rule out isolated or continuous degeneracies between sampled points and therefore does not establish continuous global injectivity. For points i, j, normalized parameter separation is

$$d_\theta(i,j) = \sqrt{\left(\frac{k_i - k_j}{\Delta k}\right)^2 + \left(\frac{w_{q,i} - w_{q,j}}{\Delta w_q}\right)^2}, \tag{7}$$

where both spans are taken from the analysis domain. Observable differences use the nominal grid-wide q95–q05 feature scales. The primary cross-set distance is the RMS of standardized feature differences rather than a raw Euclidean norm, because the combined set has 31 dimensions while ringdown has four.

At $k = 0$, the ringdown distance across all 55 nominal-w_q pairs is exactly zero. For ringdown plus photon geometry the minimum and median are zero and the maximum is 2.63×10^{-12}. This is the expected $k = 0$ null control and reproduces the exact Schwarzschild structural degeneracy to numerical precision; these pairs are not counted as undesirable collisions.

Table 10: Closest finite-k RMS standardized observable pair at each minimum normalized parameter separation.

Feature set	$d_\theta \geq 0.10$	≥ 0.25	≥ 0.50	≥ 0.75
Ringdown	0.005832	0.012668	0.025170	0.041655
Rdown + photon geom.	0.007710	0.013406	0.031550	0.072741

Table 11: Nearest-observable-neighbor parameter distance and preservation of the nearest parameter-space neighbor. Poorer exact local neighbor preservation does not imply increased distant ambiguity.

Feature set	median d_θ	p90	max.	top-1	top-3	top-5
Ringdown	0.09524	0.21513	0.23216	40.9%	71.8%	75.5%
Rdown + photon geom.	0.13171	0.21759	0.21886	20.0%	48.2%	67.3%

No nearest observable neighbor lies beyond $d_\theta = 0.25$ for either set, and no pair with $d_\theta \geq 0.25$ lies below the fifth-percentile local spacing. The closest combined pair at $d_\theta \geq 0.5$ is 12.8 times the largest measured combined-feature 161-to-321 RMS change. Photon geometry improves the

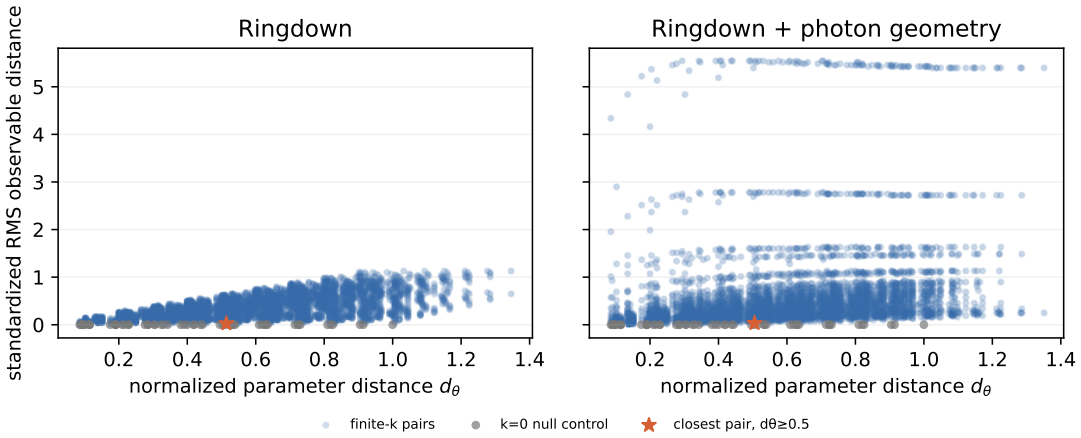


Figure 18: All-pairs finite-domain ambiguity audit. The horizontal axis is normalized parameter distance and the vertical axis is RMS standardized observable distance. Grey $k = 0$ pairs show the required structural degeneracy; distant finite-k pairs are the relevant sampled-domain diagnostic. Stars mark the closest finite-k pair with $d_\theta \geq 0.5$. This plot tests only the sampled physical grid and does not establish continuous global injectivity.

distant-pair diagnostics but not every local ordering statistic. Numerically, the closest distant-pair distance increases from 0.02517 to 0.03155 against a maximum measured forward-resolution change of 0.002462; the number of distant pairs below local-median spacing falls from 13 to 6, and the maximum nearest-observable-neighbor parameter distance changes from 0.23216 to 0.21886. Photon geometry does not preserve every local nearest-neighbor ordering even though it improves the distant-pair diagnostics.

J MLP Extrapolation-Failure Audit

The machine-readable table contains standardized input magnitude, hidden activation magnitude, iteration status, overflow flag, training distance, and resolution response for every catastrophic record. It distinguishes optimization failure from extrapolation instability without clipping.

Exact frozen-estimator shift summaries used in Fig. 9 are retained in Table 12 for numerical reference.

Table 12: Targeted 321-phase frozen-estimator robustness.

Model	median	95th percentile	max. w_q
HGB	0	0.08122	0.30593
RF	0	0.08509	0.27053
MLP	0.001143	0.008223	0.03819

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